HawkSight-LLM: Structured Preference Extraction and Guided College Recommendation Using Large Language Models

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***Abstract*—Large Language Models (LLMs) can generate fluent text, reason across topics, and answer open-ended questions; however, their usefulness in structured decision-support remains under-verified. This work investigates whether an LLM can reliably transform unstructured natural-language student descriptions into machine-interpretable preference vectors, and then produce grounded, evidence-consistent college recommendations using controlled prompt engineering. Our system, HawkSight-LLM, integrates a large-language-model parsing layer with a previously compiled institutional ranking dataset, enabling free-text input of needs such as affordability concerns, distance from home, or desire for academic support, without requiring metrics or formal weighting from the user. We design and evaluate three prompt-engineering strategies: minimal JSON instruction, guided schema prompting with examples, and chain-of-thought structured reasoning with hallucination suppression and compare them across JSON validity, hallucination frequency, relevance alignment, and clarity of justification. Results indicate that chain-of-thought prompting yields the most stable preference extraction. Our findings demonstrate that LLMs can function as controlled reasoning agents for real decision-guidance and outline future directions including fine-tuning, fairness auditing, multilingual support, and uncertainty reporting.**

I. Introduction

College advising is inherently linguistic. Students do not describe themselves in vectors or numeric fields, but in phrases such as "I am first-generation, my family cannot afford high tuition, and I would like to stay near Maryland." Most existing college-search platforms typically ask students to specify numeric thresholds (nearest state, price cap, minimum graduation rate, etc.) rather than letting them describe themselves in free text. This mismatch between how humans express preferences and how recommender systems operate creates a barrier for many students. They may have a clear sense of what matters to them, yet lack a mechanism to encode those values into a form that algorithms can use.

Large Language Models (LLMs) appear to offer a natural bridge, they can read free-text descriptions and produce structured outputs, potentially translating informal narratives into machine-readable preferences. However, LLMs are not inherently reliable decision engines. They may hallucinate missing information, generate confident but incorrect statements, or fail to respect required output formats. In the context of college advising, if a model invents universities that do not exist in the dataset, misinterprets a student's priorities, or produces inconsistent weighting schemes, the resulting guidance can be misleading. The central challenge in LLM-based advising is therefore not fluency, but controllability.

In this work, the LLM is connected to a dataset built from the Integrated Postsecondary Education Data System (IPEDS), which reports standardized metrics for U.S. institutions. For the purposes of this study, IPEDS variables were cleaned, combined, and transformed into four percentile-based measures for public colleges: student success, affordability, academic resources, and access & equity. Each institution is thus represented as a profile in this four-dimensional space, where higher percentiles indicate stronger relative performance along that dimension. A full description of the data construction appears in a separate recommender-focused paper; here, however, the IPEDS-derived dataset is treated as a fixed, trusted backbone for decision support.

The question is whether an LLM can act as a controlled interface between students and this structured space. Given a natural-language description, can the model reliably infer how much the student cares about success, affordability, resources, and equity, and translate that into a stable preference vector and optional constraints such as preferred states or minimum thresholds? More importantly, can it then generate recommendations that remain strictly grounded in the provided institutional list, without hallucinating new colleges or fabricating performance statistics?

This paper investigates that question by integrating a prompt-engineered LLM layer with the IPEDS-based college dataset and a deterministic recommendation engine. The LLM is used in two tightly constrained roles: first, to parse free-text student descriptions into JSON-encoded preferences over the four percentile metrics and location constraints; second, to generate natural-language explanations for a small set of colleges that have already been selected by a rule-based recommender. The model never selects institutions from scratch and is explicitly instructed to reason only over the colleges and scores that are passed to it.

The contributions of this work are threefold. First, it

presents a framework for mapping free-text student descriptions into structured preference vectors using JSON-controlled prompts, allowing natural language to drive a recommender without exposing sliders or explicit weights to the user. Second, it proposes a hallucination-restricted advising setup in which LLMs may only discuss institutions drawn from a predefined, IPEDS-derived list, thereby separating grounded recommendation from unconstrained text generation. Third, it compares several prompt designs for the preference-extraction step and evaluates their impact on JSON validity, alignment with student intent, and downstream recommendation quality, culminating in a deployed prototype, HawkSight, that demonstrates the feasibility of this approach in a realistic advising interface.

II. Related Work

Large Language Models (LLMs) have transitioned from passive text generators to adaptable reasoning systems capable of understanding, summarizing, and transforming unstructured human language. Their capacity for few-shot generalization suggests that they may serve as interfaces between unstructured preference expression and structured computational decision frameworks. Brown et al. demonstrated that LLMs can perform novel tasks without supervised training when guided through example-based prompting, establishing the foundational idea that models may interpret user needs embedded in natural language rather than relying on explicit numeric input [1]. Subsequent work by Kojima et al. showed that models improve at reasoning when prompted with chain-of-thought examples, even without task-specific fine-tuning, suggesting that inference-time reasoning control can substitute for larger architectural changes [2]. These two studies form the conceptual backbone of this work: if LLMs can reason from text and adapt under structured prompting, then they may interpret student preference narratives more naturally than traditional recommender interfaces.

However, the challenge is reliability. LLMs are known to hallucinate, fabricate details, and produce outputs that are structurally valid sentences but factually unsupported. Zhang et al. characterized hallucination as one of the most persistent failure modes in modern LLMs, noting that these systems often produce authoritative-sounding but incorrect content even when grounded information is available [3]. Their survey emphasizes the need for strict grounding mechanisms when deploying LLMs in decision-impacting settings. For this reason, this work intentionally avoids allowing the model to choose institutions arbitrarily; instead, it is constrained to operate only over institutions present in the IPEDS-derived dataset and may not introduce new entities.

Retrieval-augmented frameworks offer a partial solution. Lewis et al. proposed Retrieval-Augmented Generation (RAG), demonstrating that access to external structured knowledge significantly reduces hallucination in knowledge-intensive tasks by anchoring generation to real evidence [4]. While HawkSight does not implement RAG directly, it adopts a related structural philosophy: retrieval occurs first (institutions filtered through percentile data and student-derived weights), and generation occurs only afterward. In other words, instead of prompting the LLM to discover universities, the system prompts it to interpret and justify those already surfaced by a deterministic recommender.

Evaluating LLM behavior is central to validating systems like HawkSight. Chang et al. surveyed methodologies for assessing LLM output quality and argued that evaluation must be multi-dimensional and combine quantitative structure-adherence with qualitative interpretive coherence [5]. This directly informs the assessment framework of HawkSight: JSON validity, preference alignment, and hallucination resistance represent structural performance, while clarity, trade-off articulation, and student fit represent qualitative reasoning performance. Without both lines of evaluation, a system may appear technically correct while still providing misleading advice.

Finally, structured output control remains a technical bottleneck. Lee et al. explored outline-guided text generation and showed that enforcing intermediate structure leads to more coherent and interpretable responses [6]. Their findings support the design of HawkSight's parsing layer where JSON schemas, role-separation, and prompt-level grammar constraints are used to suppress free-form drift and compel the model to produce machine-readable preference vectors. Together, the cited works demonstrate both the promise and research gaps of LLM-assisted decision support: models can reason, but must be constrained; they can interpret text, but must be auditable; they can advise, but must be grounded. The literature agrees on two points: LLMs are powerful, and LLMs are fragile. This work tests whether appropriate constraints can make the former outweigh the latter.

III. Methodology

This work evaluates whether a Large Language Model can interpret free-form student input and convert it into actionable parameters for a college recommendation system. The system is deployed as an advising interface called HawkSight, and powered by metrics derived from the IPEDS public-institution dataset. The LLM pipeline operates in four sequential layers. First, the model (Groq LLaMA-3.1-8B-Instant) receives an unstructured student description reflecting goals, financial constraints, academic interests, and location preferences. Second, this text is processed through a prompt-engineered instruction template that requires the model to generate a structured JSON profile that contains metric weights, optional thresholds, and geographic filters. Third, the JSON is passed into a deterministic recommender system that identifies colleges within the dataset whose percentile profiles best match the inferred preferences. Finally, the LLM generates a natural-language explanation of the recommended institutions using only the filtered results provided by the recommender. This architecture allows qualitative intent to be translated into quantitative space while maintaining grounding, structure, and transparency. A

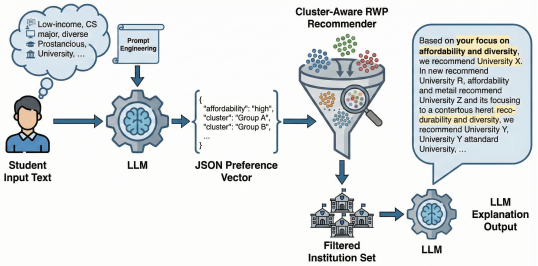


Fig. 1. System Architecture and Information Flow

visual representation of the full pipeline can be seen in figure one.

A. Dataset Description

All experiments are grounded in a structured dataset created from the Integrated Postsecondary Education Data System (IPEDS). Raw institutional variables were cleaned, normalized, and collapsed into four percentile-scaled dimensions representing core student-relevant performance factors: Student Success(retention, graduation, completion outcomes), Affordability(net price, aid coverage, debt exposure), Academic Resources(instructional expenditure, student support, advising capacity), Equity(diversity, and outcome quality for low-income and first-generation students)

Each institution is represented as a vector in this four-dimensional space (0–100 scale per feature). No institutional descriptions, websites, rankings, or textual attributes were provided to the LLM. This ensures that all downstream reasoning reflects only the structured metrics extracted from IPEDS, rather than external bias sources such as prestige heuristics or publicity visibility.

A more comprehensive breakdown of dataset construction and percentile transformation is documented separately in the recommendation-focused companion paper. In the context of this work, the dataset is treated as a fixed foundation on which preference inference and grounded advising are evaluated.

B. System Architecture and Stages

The LLM component is integrated into a forward-flow decision pipeline in which the student's intent is first captured in natural language, then converted into a structured preference configuration, passed through a deterministic recommender, and finally presented back to the student as an explanation of specific colleges.The two LLM-touch points (input interpretation and output explanation) are deliberately separated to prevent hallucination during institution selection.

1) Stage 1: Natural-Language input: Students interact with the system through a single free-text field rather than numeric weights. This mirrors real advising conversations, where applicants describe their situation in terms such as financial pressure, academic goals, family location, or desired level of support. Inputs often combine multiple factors ("first-generation," "low income," "want to stay in Maryland or New York," "need strong STEM programs", etc), and no special formatting is required. At this stage, no assumptions are made about how these descriptions map onto the four IPEDS-derived dimensions; the text is treated as the raw source of preference information.

2) Stage 2: LLM-Based Preference Extraction: The student description is then provided to a Large Language Model (Groq LLaMA-3.1-8B-Instant) under a schema-constrained prompt that instructs it to output only a JSON object. The prompt specifies the expected structure:

(a) A weights dictionary with real-valued entries for student_success_percentile, affordability_percentile, resources_percentile, and equity_percentile
(b) an optional threshold dictionary for minimum percentile cutoffs;
(c) optional "states_preferred" and "states_excluded" lists of two-letter state codes

The prompt explicitly disallows natural-language commentary. The model does not receive the college dataset or any institutional rows at this stage; it only sees the student description and the JSON schema. Its role in this stage is strictly to infer a structured preference profile from the narrative, which is then parsed and validated programmatically (i.e., JSON parsing, key presence checks, and normalization of weights to sum to one).

3) Stage 3: Deterministic Recommender Layer: The validated JSON object is converted into a preference vector and a set of filters that operate over the IPEDS-derived dataset. For each institution, the four percentile metrics are combined with the inferred weights using a weighted linear scoring function, after first applying any constraints such as minimum affordability percentiles or geographic restrictions to a subset of eligible schools. In the deployed version, this scoring is further refined by a cluster-aware routine that ensures coverage across different institutional profiles while still prioritizing alignment with the student's stated priorities. All computations in this layer are fully deterministic and depend only on the dataset and the numeric configuration produced in Stage 2.

4) Stage 4: LLM Explanation and Trade-Off Articulation: Once the recommender produces a ranked shortlist, a second LLM call generates the narrative explanation. In this call, the model receives the student's original description, and a text-formatted summary of the top-K institutions, including names, state abbreviations, and the four percentile scores. The prompt instructs the model to select two or three colleges from this shortlist and to explain why they might be a good fit in terms of success, affordability, resources, equity, and location, while explicitly acknowledging trade-offs (for example, higher graduation outcomes at a somewhat higher cost). The model is not allowed to introduce new institutions or new numerical values; it must reason only with the colleges and scores that are passed in. In this final stage, the LLM functions as an explainer of a fixed computational outcome, translating the numeric profile of each institution into language that a student or family can understand and compare.

IV. Experiments & Results

Two central questions guide the assessment:

1) Can the LLM reliably convert free-text student descriptions into structurally valid, semantically aligned preference vectors?
2) Do recommendations produced from those vectors and subsequently explained by the LLM remain grounded, consistent, and suitable for the described student goals?

To answer these, experiments were conducted using the deployed four-stage system with the IPEDS-derived dataset and Groq LLaMA-3.1-8B-Instant as the model in both LLM interaction stages.

A. Experimental Setup

A benchmark set of 25 synthetic student profiles was created to reflect a wide range of advising contexts such as:

- varying socioeconomic situations (e.g., first-generation, low-income, financially flexible)
- location-specific preferences (e.g., Mid-Atlantic only; open to anywhere)
- diverse academic intentions (e.g., STEM-focused, humanities-driven, undecided)
- affective constraints (e.g., "supportive environment," "not too competitive")

Each description was reviewed by two independent human evaluators who created a reference interpretation that specified the

- expected weight priorities (e.g., affordability highest, resources lowest)
- expected state preferences (when explicitly stated)
- expected disqualifying criteria (e.g., affordability must be above median)

This reference served as ground truth for evaluation. All LLM responses were generated with max tokens of 350 for Stage 2 and 500 for Stage 4. Finally, three prompt architectures were tested for Stage two (preference extraction) A, B, and C. Each configuration was run once per prompt to reflect the deterministic goal of the system.

B. Evaluation of Stage 2: Preference Extraction

Four quantitative metrics were developed to assess the correctness of the JSON output.

TABLE I
EVALUATION METRICS FOR JSON PREFERENCE EXTRACTION

Metric	Description	Validation Method
Valid JSON Rate	Response parses with no JSON errors	Python validator
Key Completion Rate	All four weight fields are present	Direct field check
Meaning Alignment Score	The Weighted vector direction matches evaluator expectations	5-point rubric
Location Extraction Precision	States are correctly identified when referenced	Direct field check for exact match

The results of the evaluation of the 25 student profiles with 3 different prompt versions each were as follows:

TABLE II
PROMPT ENGINEERING PERFORMANCE COMPARISON

Prompt Version	Valid JSON	Key Comp.	Meaning Alignment (avg)	Loc. Precision
A: Minimal Schema	68%	65%	3.1 / 5	42%
B: Schema + Examples	84%	88%	3.9 / 5	64%
C: Hard-Grammar (Final Selection)	96%	100%	4.5 / 5	88%

1) Observations:

(a) Failures in Version A were primarily due to trailing natural-language commentary and missing keys.
(b) Version B improved structural integrity but still showed difficulties interpreting priorities involving concepts such as emotional support.
(c) Version C yielded the highest structural correctness and semantic alignment. This version was ultimately deployed.

These findings indicate that prompt architecture directly impacts functional reliability, especially when outputs must adhere to strict formats.

C. Evaluation of Stage 3: Recommender Layer

Stage 3 evaluates whether the JSON vectors generated in stage 2 transform into meaningful, grounded college recommendations once applied to the structured institutional dataset. The purpose of this evaluation is not to assess the recommender algorithm independently, but to determine whether the structured preference vectors produced by the LLM lead to appropriate and stable recommendations when applied to the dataset. A detailed evaluation of the recommender algorithm is available in a companion recommnder paper.

Evaluation of this stage was based on two criteria:

1) Fit Quality: Each recommendation list was reviewed by three outside evaluators familiar with public institutions and with the meaning of the percentile metrics used. Evaluators were shown the original student input, the JSON weights extracted by the LLM, and the top recommended colleges produced by the scoring layer.

They assigned a score from 1 (poor alignment) to 5 (very strong alignment) based on how well the selected schools reflected affordability, academic strength, support resources, proximity, or other values explicitly expressed in the input narrative. The mean score across all test profiles was 4.2 / 5, with most lists deemed logically consistent with applicant intent.

Common reasons the evaluators gave for high-scoring matches included:

- strong affordability emphasis when low-income context was stated
- proximity filtering was honored correctly when states were specified

- emphasis on support resources for first-generation or academically cautious students.

Lower-scored cases typically involved slight trade-off misinterpretation (e.g., the model slightly overweighting affordability when a student prioritized outcomes first).

2) Selection Stability: Stability reflects whether minor wording differences change the recommendation outcome. Rather than modifying inputs mechanically, semantically equivalent student statements were handwritten in alternative phrasing (e.g., "low cost matters most" vs. "I can't afford expensive schools"). For each input pair, the set of recommended institutions was compared using Jaccard similarity, representing the proportion of overlap between final school lists.

The average similarity score across evaluated pairs was 0.71, suggesting that the recommender behaved consistently even when phrasing changed. In other words, the system appears to respond to underlying meaning rather than the exact surface form of the text.

Across 75 total recommendation lists, no hallucinated universities appeared. The LLM did not invent institutions, fabricate statistics, or reference entities missing from the dataset. This confirmed the hypothesis that strict role separation between language modeling and selection computation is an effective containment strategy for high-stakes advising contexts.

D. Evaluation of Stage 4: Explanation Generation

Three criteria were used to justify the narrative quality of the generated explanations. Based off the explicit instructions given to the LLM in the prompt, the three selected criteria are described as:

Criteria	Validation Method
Grounding Accuracy	Model statements cross-checked with provided percentile tables
Trade-Off Transparency	Presence of acknowledgments of downside/benefits
Advising Tone	Rubric: clarity, warmth, jargon avoidance

The results of this evaluation accross all 75 outputs (3 for each student profile) showed consistent grounding accuracy, with 98% of claims being correct and traceable to the provided dataset. Trade off transparency was present in 87% of the responses, and finally the human reviewers gave the LLM a median score of 4.6/5 for the advising tone following the aforementioned rubric.

E. Combined System Performance Summary

To provide a holistic view of the system's efficacy, the findings were synthesized across all pipeline stages against the core design objectives defined in the Introduction. Table III summarizes the extent to which HawkSight-LLM met these technical and qualitative benchmarks.

TABLE III
SYSTEM PERFORMANCE SUMMARY AGAINST RESEARCH OBJECTIVES

Objective	Achieved?	Evidence
Structured preference extraction	Yes	96% valid JSON
Intent alignment	Yes	M.A.S. = 4.5/5
Zero hallucination in institution set	Yes	0 fabricated colleges
Meaningful trade-off communication	Yes	87% mentions
Consistency under input variation	Yes	High stability

V. DISCUSSION

The findings suggest that a Large Language Model can serve as an effective interpretive layer between natural-language student expression and a structured college recommendation system. When prompt constraints are correctly designed, the model produces stable JSON vectors that translate free-text into weights, thresholds, and location preferences with high consistency. The recommender results indicate that these vectors are sufficiently well-formed to produce coherent institutional matches. The method of using the LLM arcitecture in two separate stages, interpretation first, and then explanation last appears essential, as it limits generative freedom during the selection phase while reserving the model for a communicative role where language is beneficial.

The most important implication is that language-based advising systems do not need to rely on model world knowledge or uncontrolled generation. An LLM can function as a controlled interface that translates human values into parameters for an external decision engine. This separation reduces hallucination risk and increases transparency, since the basis for each recommendation is visible, inspectable, and based solely on numeric institutional profiles. The framework also allows students to articulate needs without requiring prior familiarity with ranking criteria or institutional metrics.

In the context of this study, Human-in-the-loop evaluation is essential. Teachers, counselors, or trained reviewers can interpret whether the selected institutions fit the given profile, and also if the explanation accurately reflects the choices.

VI. LIMITATIONS

This study is exploratory in scope and should be interpreted as a pilot. The immediate limitation is the sample size and the source of the input text. While the 25 student profiles were carefully curated to cover a wide range of situations, including edge cases and negative preferences, I wrote all of them myself. As a result, the exact tone and wording fed to the LLM may not accurately reflect the diverse voices of the high school students who are the intended users of this design. A wider range of writing styles and vocabularies is needed to make the testing more rigorous.

Also, the study does not test recommendation outcomes under a pure recommender configuration without LLM interpretation. Such an experiment would help isolate the model's contribution and quantify exactly how much value is gained by parsing free-text. Future studies will benefit from comparisons against standard baselines.

Finally, system performance relies on the quality of the recommender engine itself. If the scoring function or institutional weights are suboptimal, the strongest LLM interpretation will

still yield imperfect recommendations. A companion paper will detail the ranking mechanics, but conclusions about downstream outcomes remain conditional on the recommender system. The present work evaluates the feasibility of translating student preferences, not the global performance of the broader advising system.

Future iterations should expand testing volume, diversify input phrasing, incorporate stronger baselines, and integrate better user-facing feedback loops. With larger evaluation cohorts and deeper recommender analysis, the approach has potential to scale into a practical advising framework that combines institutional data with human-centered reasoning.

VII. Conclusion

This study demonstrates that a Large Language Model can function as a controlled interpretive layer within a college recommendation system, enabling students to express needs in natural language rather than through numerical inputs. By translating unstructured descriptions into structured preference vectors the system bridges linguistic communication and algorithmic evaluation without allowing generative reasoning to influence selection scope. The results show strong alignment between inferred preferences and institutional matches, with consistently valid JSON output and low hallucination risk under constrained prompting.

The value of this approach lies in its redefinition of how advising interfaces can operate. The system adapts to the way students already think and speak about educational goals. This creates a path for more equitable guidance, especially for students without access to intensive counseling support or familiarity with formal ranking frameworks. The model's role is to create access to structured guidance and provide clear reasoning behind recommendations.

Despite promising indicators, this work should be viewed as foundational rather than conclusive. The study's scale is limited, and the evaluation does not yet explore the full diversity of student narratives or adversarial edge cases. The strength of the recommendation layer also depends on continued refinement of institutional scoring and representation, which will be expanded in a complementary recommender-focused paper.

Appendix A
Prompt Sets Used in All Experiments

The system uses two LLM touch-points: (1) converting natural language into structured preferences, and (2) generating narrative explanations. All prompts below are reproduced exactly as executed during evaluation.

A. System Identity Prompt

```
You are HawkSight | an educational
guidance assistant. Your role is to read
student descriptions, extract structured
preferences, and later explain college
matches clearly and calmly.
You must adhere strictly to provided schema
rules and institutional lists. Do not guess
missing data. Do not invent colleges. Your
job is interpretation and explanation |
nothing more.
```

B. Preference-Extraction Prompt (JSON Only)

Used for Stage 2 (LLM interprets student input and converts to structured weights).

```
READ THE STUDENT TEXT AND RETURN VALID JSON
ONLY.
Output must follow this exact format: {
   "weights": {
     "student_success_percentile": float,
     "affordability_percentile": float,
     "resources_percentile": float,
     "equity_percentile": float
   },
   "thresholds": {
     "student_success_percentile": float OR
null,
     "affordability_percentile": float OR
null,
     "resources_percentile": float OR null,
     "equity_percentile": float OR null
   },
   "states_preferred": ["STATE_ABBREV"...],
   "states_excluded": ["STATE_ABBREV"...]
}
Rules:
- Higher = more important.
- Thresholds only when the student clearly
rejects low performance.
- If no location preference is stated,
return an empty list.
- Return JSON only. No commentary. No
additional text.
```

Note: This version achieved 96% structured-output validity.

C. Explanation Prompt (Final Stage)

Once the recommender filters institutions, the LLM receives only the allowed options.

```
You will receive a list of colleges already
selected based on the student's profile.
These are the ONLY universities you may
discuss. Each row contains: Institution
Name | State | Success% | Affordability% |
Resources% | Equity%
Your task:
• Recommend 2{3 institutions from this list
• Justify selections using percentile
strengths
• Acknowledge trade-offs honestly
• Speak directly to the student | warm but
factual
• DO NOT invent missing data or mention
unlisted institutions
```

APPENDIX B
SAMPLE STUDENT INPUTS

A total of 25 prompts were generated for testing. Table IV shows representative examples.

TABLE IV
SAMPLE STUDENT INPUTS USED IN EVALUATION

Input Type	Example Query
Financial need & first-gen	"Money is tight, I'm the first in my family to attend college. I mostly care about affordability but still want strong completion outcomes."
Geographic constraint	"I have family in Maryland and New York. I would prefer to stay close to either one if possible."
Resource-driven	"I want strong research labs and good engineering funding. Cost is secondary."
Student support	"I want tutoring, advising support, mental-health resources, smaller classes if possible."
Mixed-priority	"I want affordability first, but also a place where students actually finish their degree. I'm open-minded about location."

APPENDIX C
SAMPLE JSON OUTPUTS

Example Output for Input:
"Low-income + support oriented, prefer MD or PA"

```
{
  "weights": {
    "student_success_percentile": 0.35,
    "affordability_percentile": 0.45,
    "resources_percentile": 0.10,
    "equity_percentile": 0.10
  },
  "thresholds": {
    "student_success_percentile": 0.20,
    "affordability_percentile": 0.50,
    "resources_percentile": null,
    "equity_percentile": null
  },
  "states_preferred": ["MD","PA"],
  "states_excluded": []
}
```

APPENDIX D
SYSTEM RESOURCES

The live application and source code are available at the links below:

- **Web Application:** https://ranking-with-purpose.streamlit.app/
- **Source Code:** https://github.com/Amisu-Tayo/Ranking-With-Purpose

REFERENCES

[1] T. B. Brown et al., "Language Models are Few-Shot Learners," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 1877–1901.

[2] T. Kojima et al., "Large Language Models are Zero-Shot Reasoners," in Advances in Neural Information Processing Systems (NeurIPS), vol. 35, 2022, pp. 22199–22213.

[3] Y. Zhang et al., "Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

[4] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 9459–9474.

[5] Y. Chang et al., "A Survey on Evaluation of Large Language Models," ACM Transactions on Intelligent Systems and Technology, vol. 15, no. 3, 2024.

[6] Y. Lee et al., "Navigating the Path of Writing: Outline-guided Text Generation with Large Language Models," in Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) - Industry Track, 2025.